Module 6: Genetics, evolution and ecosystems

This module covers the role of genes in regulating and controlling cell function and development. Heredity and the mechanisms of evolution and speciation are also covered.

Some of the practical techniques used to manipulate DNA such as sequencing and amplification are considered and their therapeutic medical use. The use of microorganisms in biotechnology is also covered. Both of these have associated ethical considerations and it is important that learners develop a balanced understanding of such issues.

Learners gain an appreciation of the role of microorganisms in recycling materials within the environment and maintaining balance within ecosystems. The need to conserve environmental resources in a sustainable fashion is considered, whilst appreciating the potential conflict arising from the needs of an increasing human population. Learners also consider the impacts of human activities on the natural environment and biodiversity.

Learners are expected to apply knowledge, understanding and other skills developed in this module to new situations and/or to solve related problems.

6.1 Genetics and evolution

6.1.1 Cellular control

The way in which cells control metabolic reactions determines how organisms, grow, develop and function.

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	types of gene mutations and their possible effects on protein production and function	To include substitution, insertion or deletion of one or more nucleotides **AND** the possible effects of these gene mutations (i.e. beneficial, neutral or harmful).

(b)	the regulatory mechanisms that control gene expression at the transcriptional level, post-transcriptional level and post-translational level	To include control at the, • transcriptional level: *lac* operon, and transcription factors in eukaryotes. • post-transcriptional level: the editing of primary mRNA and the removal of introns to produce mature mRNA. • post-translational level: the activation of proteins by cyclic AMP. HSW2
(c)	the genetic control of the development of body plans in different organisms	Homeobox gene sequences in plants, animals and fungi are similar and highly conserved **AND** the role of Hox genes in controlling body plan development. HSW7
(d)	the importance of mitosis and apoptosis as mechanisms controlling the development of body form.	To include an appreciation that the genes which regulate the cell cycle and apoptosis are able to respond to internal and external cell stimuli e.g. stress.